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Introduction

Technology development as the time goes by, creating an impact to future businesses and companies by a strong network as the main step to a successful building process. What will come next for Gulf United Bank (GUB) if they plan to move their central building to Manama and to first settle in Riffa and Muharraq is to build and implement a robust network infrastructure capable of supporting the growing banking system. The galaxy-wide infrastructure will be tied for the websites form a centralized united system which is displayed in unified database. This means that one can expect smooth running of the facilities as users will utilize only one login information to access several services. The underlying network architecture will principally focus on scaling, reliability and durability and security to make sure speedy and secured connectivity.

In this report, we aim to present the GUB envisioned network architecture which is a modern stack comprising the most recent technologies as well as best security practices. Such a schedule which is designed in a hierarchical manner, encompassing the details from individual devices to the connections between the central office and the branches as a whole. The first task will be to make it secure enough and prioritize the backup things for the system so, it will be running 24/7.

Designing a private IPv4 addressing

City	Staff
Manama	280 Staff
	35 Management, 40 Administration, 45 marketing, 20 Human Resources, 65 Finance, 35 IT, 20 Loan services, 20 Legal.
Riffa	126 Staff
	10 Management, 20 Administration, 10 marketing, 8 Human Resources, 20 Finance, 15 IT, 33 Loan services, 10 Legal.
Muharraq	133 Staff
	10 Management, 15 Administration, 10 marketing, 8 Human Resources, 15 Finance, 15 IT, 55 Loan services, 5 Legal.

Subnet size will be based on each branch's need.

Gulf United Bank (GUB) expects Muharraq and Riffa branches staff to increase within the next 5 years so we will assign them subnets that can accommodate double the current number of employees.

City	Size	Address	Assignable Range
Manama	280	172.16.0.0/16	172.16.0.1 – 172.16.255.254
Muharraq	266 (133x2)	10.30.0.0/16	10.30.0.1-10.30.255.254
Riffa	252 (126x2)	10.20.0.0/16	10.20.0.1- 10.20.255.254
WAN		192.168.5.0/24	192.168.5.1 - 192.168.5.254

ISP		193.200.35.0/27	193.200.35.1 - 193.200.35.30
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Each Department will have its separate VLAN

Department	VLAN ID
Management	10
Administration	20
Marketing	30
Human Resources	40
Finance	50
IT	60
Loan services	70
Legal	80
Wireless*	90
Native	100

*Exclusive to Manama Branch

Manama **VLSM** 172.16.0.0/16

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range	Broadcast
50 Finance	65	126	172.16.0.0	/25	255.255.255.128	172.16.0.1 - 172.16.0.126	172.16.0.127
30 Marketing	45	62	172.16.0.128	/26	255.255.255.192	172.16.0.129 - 172.16.0.190	172.16.0.191
20 Administrator	40	62	172.16.0.192	/26	255.255.255.192	172.16.0.193 - 172.16.0.254	172.16.0.255
60 IT	35	62	172.16.1.0	/26	255.255.255.192	172.16.1.1 - 172.16.1.62	172.16.1.63
10 Management	35	62	172.16.1.64	/26	255.255.255.192	172.16.1.65 - 172.16.1.126	172.16.1.127
40 HR	20	30	172.16.1.128	/27	255.255.255.224	172.16.1.129 - 172.16.1.158	172.16.1.159
70 Loan	20	30	172.16.1.160	/27	255.255.255.224	172.16.1.161 - 172.16.1.190	172.16.1.191
80 Legal	20	30	172.16.1.192	/27	255.255.255.224	172.16.1.193 - 172.16.1.222	172.16.1.223
100 Native	8	14	172.16.1.224	/28	255.255.255.240	172.16.1.225 - 172.16.1.238	172.16.1.239
90 Wireless	8	14	172.16.1.240	/28	255.255.255.240	172.16.1.241 - 172.16.1.254	172.16.1.255
R1	2	2	172.16.2.16	/30	255.255.255.252	172.16.2.17 - 172.16.2.18	172.16.2.19
R2	2	2	172.16.2.20	/30	255.255.255.252	172.16.2.21 - 172.16.2.22	172.16.2.23
Servers	3	6	172.16.2.24	/29	255.255.255.248	172.16.2.25 - 172.16.2.30	172.16.2.31

Manana IP Assignments

Device	Interface	IP Address	Default Gateway	Notes
Manama_Router	S0/0/0 (DCE)	172.16.2.17/30	N/A	
	S0/0/1 (DCE)	172.16.2.21/30	N/A	
	S0/1/1 (DCE)	193.200.35.2/29	N/A	
	S0/2/0 (DCE)	192.168.5.6/30	N/A	
	S0/2/1 (DCE)	192.168.5.2/30	N/A	
	G0/0	172.16.2.25/29	N/A	
ISP	S0/1/1	193.200.35.17/30	N/A	
MN_R1	S0/0/0	172.16.2.18/30	N/A	
MN_R2	S0/0/1	172.16.2.22/30	N/A	
MN_MNG_PC	F0/23	172.16.1.65/26	172.16.1.126	Management (VLAN 10)
MN_ADMIN_PC	F0/24	172.16.0.193/26	172.16.0.254	Administration (VLAN 20)
MN_MKT_PC	F0/23	172.16.0.129/26	172.16.0.190	Marketing (VLAN 30)
MN_HR_PC	F0/24	172.16.1.129/27	172.16.1.158	HR (VLAN 40)
MN_FIN_PC	F0/23	172.16.0.1/25	172.16.0.126	Finance (VLAN 50)
MN_IT_PC	F0/24	172.16.1.1/26	172.16.1.62	IT (VLAN 60)
MN_LOAN_PC	F0/23	172.16.1.161/27	172.16.1.190	Loan (VLAN 70)
MN_LEGAL_PC	F0/24	172.16.1.193/27	172.16.1.222	Legal (VLAN 80)
Wireless	F0/1	172.16.1.244/28	172.16.1.240	Wireless (VLAN 90)
Inside network		10.10.10.1/24		
MN_LAPTOP		10.10.10.101/24	10.10.10.1	Wireless (VLAN 90)
SERVER_1(Web)		172.16.2.26/29	172.16.2.25	
SERVER_2(Email)		172.16.2.27/29	172.16.2.25	

Muharraq **VLSM** 10.30.0.0/26

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range	Broadcast
70 Loan	110	126	10.30.0.0	/25	255.255.255.128	10.30.0.1 - 10.30.0.126	10.30.0.127
60 IT	35	62	10.30.0.128	/26	255.255.255.192	10.30.0.129 - 10.30.0.190	10.30.0.191
20 Administrator	30	30	10.30.0.192	/27	255.255.255.224	10.30.0.193 - 10.30.0.222	10.30.0.223
50 Finance	30	30	10.30.0.224	/27	255.255.255.224	10.30.0.225 - 10.30.0.254	10.30.0.255
10 Management	20	30	10.30.1.0	/27	255.255.255.224	10.30.1.1 - 10.30.1.30	10.30.1.31
30 Marketing	20	30	10.30.1.32	/27	255.255.255.224	10.30.1.33 - 10.30.1.62	10.30.1.63
40 HR	16	30	10.30.1.64	/27	255.255.255.224	10.30.1.65 - 10.30.1.94	10.30.1.95
80 Legal	10	14	10.30.1.96	/28	255.255.255.240	10.30.1.97 - 10.30.1.110	10.30.1.111
100 Native	8	14	10.30.1.112	/28	255.255.255.240	10.30.1.113 - 10.30.1.126	10.30.1.127
R1	2	2	10.30.1.112	/30	255.255.255.252	10.30.1.113 - 10.30.1.114	10.30.1.115
R2	2	2	10.30.1.116	/30	255.255.255.252	10.30.1.117 - 10.30.1.118	10.30.1.119

Muharraq IP Assignments

Device	Interface	IP Address	Default Gateway	Notes
Muharraq_Router	S0/0/0 (DCE)	10.30.1.129/30	N/A	
	S0/0/1 (DCE)	10.30.1.133/30	N/A	
	S0/1/0 (DCE)	192.168.5.26/30	N/A	
	S0/1/1 (DCE)	192.168.5.9/30	N/A	
R1_MU	S0/0/0	10.30.1.130/30	N/A	
R2_MU	S0/0/1	10.30.1.134/30	N/A	
MU_MNG_PC	F0/23	10.30.1.1/27	10.30.1.30	Management (VLAN 10)
MU_ADMIN_PC	F0/24	10.30.0.193/27	10.30.0.222	Administration (VLAN 20)
MU_MKT_PC	F0/23	10.30.1.33/27	10.30.1.62	Marketing (VLAN 30)
MU_HR_PC	F0/24	10.30.1.65/27	10.30.1.94	HR (VLAN 40)
MU_FIN_PC	F0/23	10.30.0.225/27	10.30.0.254	Finance (VLAN 50)
MU_IT_PC	F0/24	10.30.0.129/26	10.30.0.190	IT (VLAN 60)
MU_LOAN_PC	F0/23	10.30.0.1/25	10.30.0.126	Loan (VLAN 70)
MU_LEGAL_PC	F0/24	10.30.1.97/28	10.30.1.110	Legal (VLAN 80)

Riffa VLSM 10.20.0.0/16

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range	Broadcast
70 Loan	66	126	10.20.0.0	/25	255.255.255.128	10.20.0.1 - 10.20.0.126	10.20.0.127
20 Administrator	40	62	10.20.0.128	/26	255.255.255.192	10.20.0.129 - 10.20.0.190	10.20.0.191
50 Finance	40	62	10.20.0.192	/26	255.255.255.192	10.20.0.193 - 10.20.0.254	10.20.0.255
60 IT	30	30	10.20.1.0	/27	255.255.255.224	10.20.1.1 - 10.20.1.30	10.20.1.31
10 Management	20	30	10.20.1.32	/27	255.255.255.224	10.20.1.33 - 10.20.1.62	10.20.1.63
30 Marketing	20	30	10.20.1.64	/27	255.255.255.224	10.20.1.65 - 10.20.1.94	10.20.1.95
80 Legal	20	30	10.20.1.96	/27	255.255.255.224	10.20.1.97 - 10.20.1.126	10.20.1.127
40 HR	16	30	10.20.1.128	/27	255.255.255.224	10.20.1.129 - 10.20.1.158	10.20.1.159
100 Native	8	14	10.20.1.160	/28	255.255.255.240	10.20.1.161 - 10.20.1.174	10.20.1.175
R1	2	2	10.20.1.176	/30	255.255.255.252	10.20.1.177 - 10.20.1.178	10.20.1.179
R2	2	2	10.20.1.180	/30	255.255.255.252	10.20.1.181 - 10.20.1.182	10.20.1.183

Riffa IP Assignments

Device	Interface	IP Address	Default Gateway	Notes
Riffa_Router	S0/0/0 (DCE)	10.20.1.177/30	N/A	
	S0/0/1 (DCE)	10.20.1.181/30	N/A	
	S0/1/0 (DCE)	192.168.5.29/30	N/A	
	S0/1/1 (DCE)	192.168.5.34/30	N/A	
R1_RF	S0/0/0	10.20.1.178/30	N/A	
R2_RF	S0/0/1	10.20.1.182/30	N/A	
RF_MNG_PC	F0/23	10.20.1.33/27	10.20.1.62	Management (VLAN 10)
RF_ADMIN_PC	F0/24	10.20.0.129/26	10.20.0.190	Administration (VLAN 20)
RF_MKT_PC	F0/23	10.20.1.65/27	10.20.1.94	Marketing (VLAN 30)
RF_HR_PC	F0/24	10.20.1.129/27	10.20.1.158	HR (VLAN 40)
RF_FIN_PC	F0/23	10.20.0.193/26	10.20.0.254	Finance (VLAN 50)
RF_IT_PC	F0/24	10.20.1.1/27	10.20.1.30	IT (VLAN 60)
RF_LOAN_PC	F0/23	10.20.0.1/25	10.20.0.126	Loan (VLAN 70)
RF_LEGAL_PC	F0/24	10.20.1.97/27	10.20.1.126	Legal (VLAN 80)

NATIVE VLAN (VLAN 100) IP address

10.20.1.60/28 255.255.255.240 (Riffa)

Range:10.20.1.161-10.20.1.174

Switch Name	Native VLAN IP	Default gateway
S1-RF	10.20.1.174	10.20.1.177
S2-RF	10.20.1.173	
S3-RF	10.20.1.172	
S4-RF	10.20.1.171	
S5-RF	10.20.1.170	
S6-RF	10.20.1.169	

10.30.1.112/28 255.255.255.240 (Muharraq)

Range:10.30.1.113-10.30.1.126

Switch Name	Native VLAN IP	Default gateway
S1-MU	10.30.1.126	10.30.1.120
S2-MU	10.30.1.125	
S3-MU	10.30.1.124	
S4-MU	10.30.1.123	
S5-MU	10.30.1.122	
S6-MU	10.30.1.121	

172.16.1.224/28 255.255.255.240 (Manama)

Range:172.16.1.225-172.16.1.238

Switch Name	Native VLAN IP	Default gateway
S1-MN	172.16.1.238	172.16.1.230
S2-MN	172.16.1.237	
S3-MN	172.16.1.236	
S4-MN	172.16.1.235	
S5-MN	172.16.1.234	
S6-MN	172.16.1.233	

Switchs and Routers pass:

Cisco

WAN VLSM 192.168.5.0/24

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range	Broadcast
Manama-R3	2	2	192.168.5.0	/30	255.255.255.252	192.168.5.1 - 192.168.5.2	192.168.5.3
Manama-R2	2	2	192.168.5.4	/30	255.255.255.252	192.168.5.5 - 192.168.5.6	192.168.5.7
Muharraq-R2	2	2	192.168.5.8	/30	255.255.255.252	192.168.5.9 - 192.168.5.10	192.168.5.11
R1	2	2	192.168.5.12	/30	255.255.255.252	192.168.5.13 - 192.168.5.14	192.168.5.15
R2	2	2	192.168.5.16	/30	255.255.255.252	192.168.5.17 - 192.168.5.18	192.168.5.19
R3	2	2	192.168.5.20	/30	255.255.255.252	192.168.5.21 - 192.168.5.22	192.168.5.23
Muharraq-R3	2	2	192.168.5.24	/30	255.255.255.252	192.168.5.25 - 192.168.5.26	192.168.5.27
Riffa-R1	2	2	192.168.5.28	/30	255.255.255.252	192.168.5.29 - 192.168.5.30	192.168.5.31
Riffa-R3	2	2	192.168.5.32	/30	255.255.255.252	192.168.5.33 - 192.168.5.34	192.168.5.35

WAN IP Assignments

Device	Interface	Ip address	Network address
R1	S0/0/0	192.168.5.13/30	192.168.5.12
	S0/0/1	192.168.5.21/30	192.168.5.20
R2	S0/0/0	192.168.5.14/30	192.168.5.12
	S0/1/0	192.168.5.17/30	192.168.5.16
R3	S0/0/1	192.168.5.22/30	192.168.5.20
	S0/1/0	192.168.5.18/30	192.168.5.16

Device	Interfaces	Ip address	Network address
Manama R2	S 0/2/0	192.168.5.6/30	192.168.5.4
	S 0/0/1	192.168.5.5/30	
Manama R3	S 0/2/1	192.168.5.2/30	192.168.5.0
	S 0/2/1	192.168.5.1/30	
Muharraq R2	S 0/1/1	192.168.5.9/30	192.168.5.8
	S 0/1/1	192.168.5.10/30	
Muharraq R3	S 0/1/0	192.168.5.26/30	192.168.5.24
	S 0/0/0	192.168.5.25/30	
Riffa R1	S 0/1/0	192.168.5.29/30	192.168.5.28
	S 0/1/0	192.168.5.30/30	
Riffa R3	S 0/1/1	192.168.5.34/30	192.168.5.32
	S 0/1/1	192.168.5.33/30	

ISP +Manama VLSM 193.200.35.0/27

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range	Broadcast
NAT+PAT	6	6	192.168.5.0	/29	255.255.255.248	192.168.5.1 - 192.168.5.6	192.168.5.7
PPPoE	6	6	192.168.5.8	/29	255.255.255.248	192.168.5.9-192.168.5.14	192.168.5.15
ISP - Manama	2	2	192.168.5.16	/30	255.255.255.252	192.168.5.17 - 192.168.5.18	192.168.5.17

ISP +Manama IP Assignment

Device	Interface	IP address	Network address
ISP	S0/1/1	193.200.35.18/30	193.200.35.16
	G0/0	193.200.35.9/29	192.168.5.8
Manama Router	S0/1/1	193.200.35.17/30	193.200.35.16

Selecting An Appropriate Routing Protocol

Gulf United Bank (GUB) requires a routing protocol that prioritizes, scalability, redundancy, availability, and security.

We chose OSPF (Open Shortest Path First) because it is a classless routing protocol, it offers the required scalability as well as fast convergence which will ensure minimum disruption, OSPF supports large networks and dynamically adapts to changes in the network. OSPF also supports VLSM which is one of the requirements given by GUB.

```
Riffa_Router
Physical Config CLI Attributes
switchport mode access
!
interface FastEthernet0/2/3
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
router ospf 10
log-adjacency-changes
network 192.168.5.28 0.0.0.3 area 0
network 192.168.5.32 0.0.0.3 area 0
network 10.20.1.176 0.0.0.3 area 0
network 10.20.1.180 0.0.0.3 area 0
```

```
RF_R2
Physical Config CLI Attributes
no ip address
shutdown
!
router ospf 10
log-adjacency-changes
network 10.20.1.180 0.0.0.3 area 0
network 10.20.0.0 0.0.255.255 area 0
network 10.20.0.0 0.0.0.127 area 0
network 10.20.0.128 0.0.0.63 area 0
network 10.20.0.192 0.0.0.63 area 0
network 10.20.1.0 0.0.0.63 area 0
network 10.20.1.64 0.0.0.31 area 0
network 10.20.1.96 0.0.0.31 area 0
network 10.20.1.128 0.0.0.31 area 0
network 10.20.1.160 0.0.0.15 area 0
!
```

```
RF_R1
Physical Config CLI Attributes
no ip address
shutdown
!
router ospf 10
log-adjacency-changes
network 10.20.1.176 0.0.0.3 area 0
network 10.20.0.0 0.0.255.255 area 0
network 10.20.0.0 0.0.0.127 area 0
network 10.20.0.128 0.0.0.63 area 0
network 10.20.0.192 0.0.0.63 area 0
network 10.20.1.0 0.0.0.63 area 0
network 10.20.1.64 0.0.0.31 area 0
network 10.20.1.96 0.0.0.31 area 0
network 10.20.1.128 0.0.0.31 area 0
network 10.20.1.160 0.0.0.15 area 0
```

```

interface Serial0/2/0
  no ip address
  clock rate 2000000
  shutdown
!
interface Serial0/2/1
  ip address 192.168.5.1 255.255.255.252
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 10
  log-adjacency-changes
  network 192.168.5.20 0.0.0.3 area 0
  network 192.168.5.16 0.0.0.3 area 0
  network 192.168.5.0 0.0.0.3 area 0
  network 192.168.5.24 0.0.0.3 area 0
  network 192.168.5.32 0.0.0.3 area 0
!

```

```

!
interface FastEthernet0/2/2
  switchport mode access
!
interface FastEthernet0/2/3
  switchport mode access
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 10
  log-adjacency-changes
  network 192.168.5.12 0.0.0.3 area 0
  network 192.168.5.16 0.0.0.3 area 0
  network 192.168.5.4 0.0.0.3 area 0
  network 192.168.5.8 0.0.0.3 area 0
!

```

```

  switchport mode access
!
interface FastEthernet0/2/2
  switchport mode access
!
interface FastEthernet0/2/3
  switchport mode access
!
interface Vlan1
  no ip address
  shutdown
!
router ospf 10
  log-adjacency-changes
  network 192.168.5.12 0.0.0.3 area 0
  network 192.168.5.20 0.0.0.3 area 0
  network 192.168.5.28 0.0.0.3 area 0
!

```

Muharraq_Router

Physical Config CLI Attributes

```
switchport mode access
!
interface FastEthernet0/2/3
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
router ospf 10
log-adjacency-changes
network 192.168.5.8 0.0.0.3 area 0
network 192.168.5.24 0.0.0.3 area 0
network 10.30.1.128 0.0.0.3 area 0
network 10.30.1.132 0.0.0.3 area 0
.
```

MU_R2

Physical Config CLI Attributes

```
shutdown
!
router ospf 10
log-adjacency-changes
network 10.30.1.132 0.0.0.3 area 0
network 10.30.0.0 0.0.255.255 area 0
network 10.30.0.0 0.0.0.127 area 0
network 10.30.0.128 0.0.0.63 area 0
network 10.30.0.192 0.0.0.31 area 0
network 10.30.1.224 0.0.0.31 area 0
network 10.30.1.0 0.0.0.31 area 0
network 10.30.1.32 0.0.0.31 area 0
network 10.30.1.64 0.0.0.31 area 0
network 10.30.1.96 0.0.0.15 area 0
network 10.30.1.112 0.0.0.15 area 0
.
```

MU_R1

Physical Config CLI Attributes

```
shutdown
!
router ospf 10
log-adjacency-changes
network 10.30.1.128 0.0.0.3 area 0
network 10.30.0.0 0.0.255.255 area 0
network 10.30.0.0 0.0.0.127 area 0
network 10.30.0.128 0.0.0.63 area 0
network 10.30.0.192 0.0.0.31 area 0
network 10.30.1.224 0.0.0.31 area 0
network 10.30.1.0 0.0.0.31 area 0
network 10.30.1.32 0.0.0.31 area 0
network 10.30.1.64 0.0.0.31 area 0
network 10.30.1.96 0.0.0.15 area 0
network 10.30.1.112 0.0.0.15 area 0
.
```

Manama_Router

Physical Config CLI Attributes

```
shutdown
!
router ospf 10
 log-adjacency-changes
 network 172.16.2.16 0.0.0.3 area 0
 network 172.16.2.20 0.0.0.3 area 0
 network 192.168.5.0 0.0.0.3 area 0
 network 192.168.5.4 0.0.0.3 area 0
 network 193.200.35.16 0.0.0.3 area 0
 network 172.16.2.0 0.0.0.15 area 0
 network 172.16.2.24 0.0.0.7 area 0
 default-information originate
```

MN_R2

Physical Config CLI Attributes

10

```
shutdown
!
router ospf 10
 log-adjacency-changes
 network 172.16.2.20 0.0.0.3 area 0
 network 172.16.0.0 0.0.255.255 area 0
 network 172.16.0.0 0.0.0.127 area 0
 network 172.16.0.128 0.0.0.63 area 0
 network 172.16.0.192 0.0.0.63 area 0
 network 172.16.1.0 0.0.0.63 area 0
 network 172.16.1.64 0.0.0.63 area 0
 network 172.16.1.128 0.0.0.31 area 0
 network 172.16.1.160 0.0.0.31 area 0
 network 172.16.1.192 0.0.0.31 area 0
 network 172.16.1.224 0.0.0.15 area 0
 network 172.16.1.240 0.0.0.15 area 0
 network 172.16.2.0 0.0.0.15 area 0
```

MN_R1

Physical Config CLI Attributes

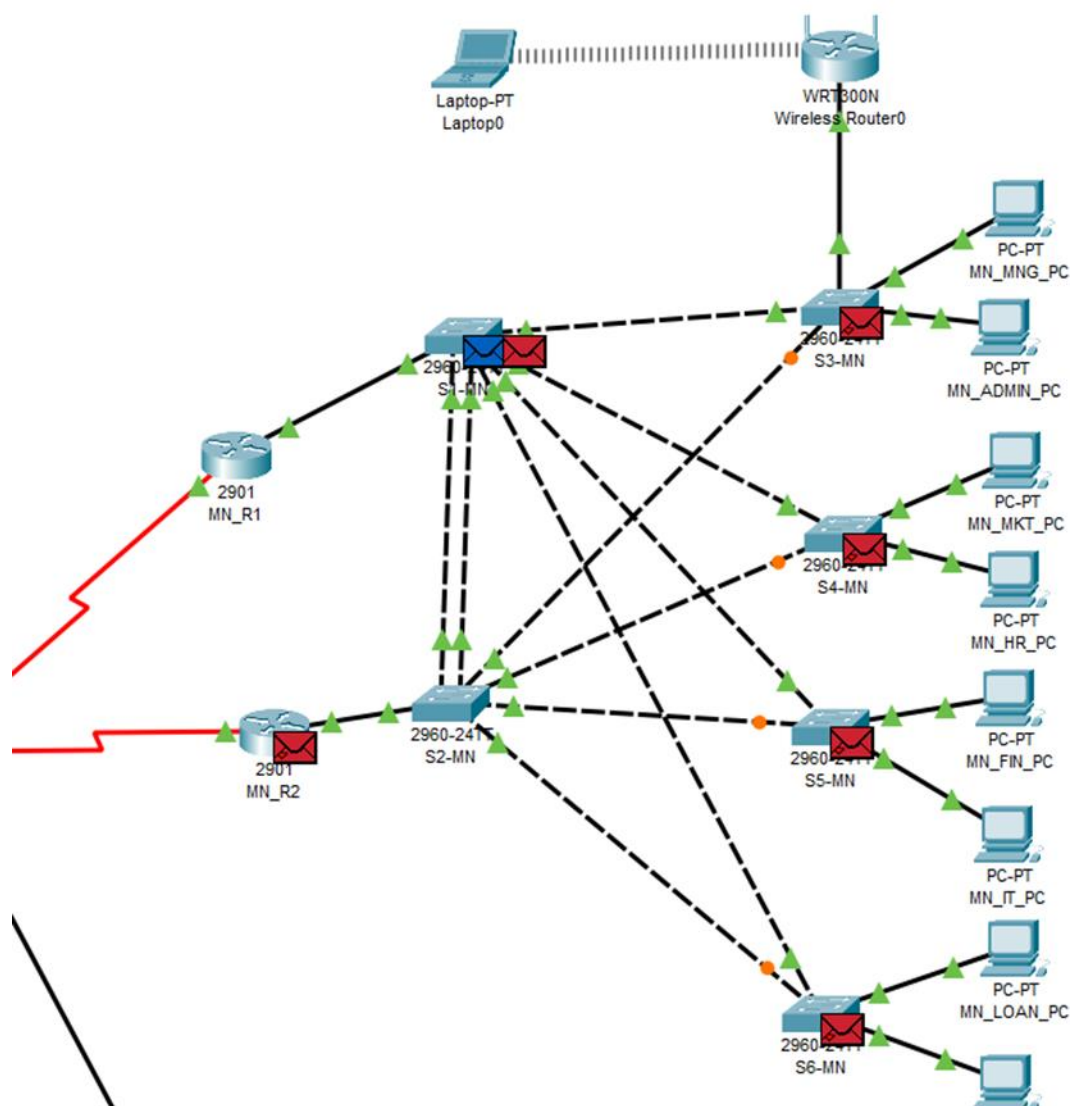
10

```
router ospf 10
 log-adjacency-changes
 network 172.16.2.16 0.0.0.3 area 0
 network 172.16.0.0 0.0.255.255 area 0
 network 172.16.0.0 0.0.0.127 area 0
 network 172.16.0.128 0.0.0.63 area 0
 network 172.16.0.192 0.0.0.63 area 0
 network 172.16.1.0 0.0.0.63 area 0
 network 172.16.1.64 0.0.0.63 area 0
 network 172.16.1.128 0.0.0.31 area 0
 network 172.16.1.160 0.0.0.31 area 0
 network 172.16.1.192 0.0.0.31 area 0
 network 172.16.1.224 0.0.0.15 area 0
 network 172.16.1.240 0.0.0.15 area 0
 network 172.16.2.0 0.0.0.15 area 0
```

Improving The Network Efficiency

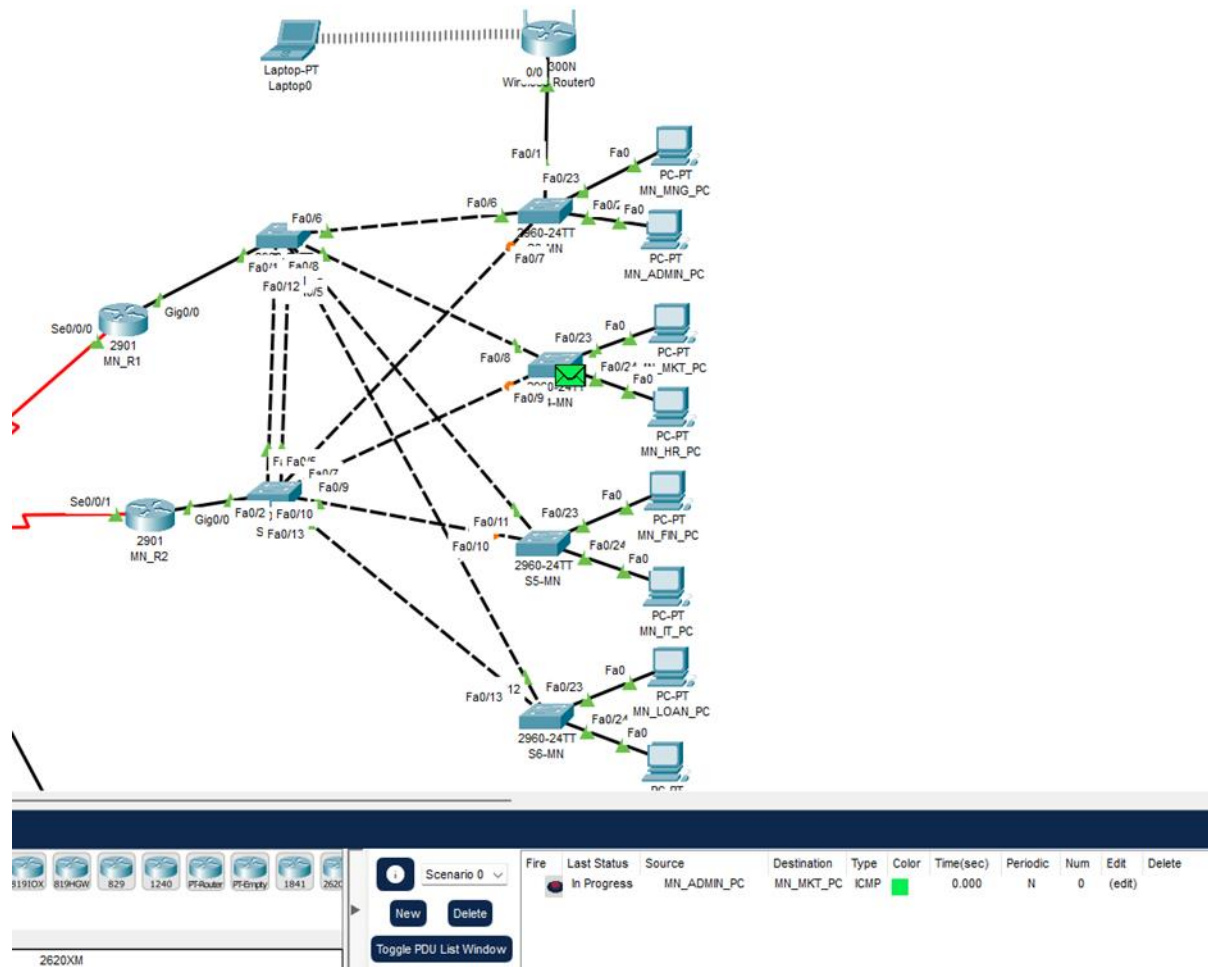
We assigned each department a VLAN to optimize the network performance, before assigning VLANs to each department, the network was slowed down by mixed traffic from all departments. Every packet was sent to every device, causing delays and high latency.

The screenshot below demonstrates the issue in Manama branch, the packet is sent to all devices which slows down the network significantly.



After Assigning a VLAN to each department, the packets were only sent to devices that are on the specific VLANs required and no more devices, this reduced traffic and latency for the whole network.

The screenshot below demonstrates the packet sent and received by the same destination; the difference is that the packet only gets sent to the relevant devices.



Security and redundancy measures

Security

-Access Layer Switches with Port Security: Port security facilitates the protection of the switch ports access by you are able to set the number of devices or MAC addresses which could be connected to a certain port. The ports security feature allows you to restrict access to your network and prevent the connection of unauthorized devices thus, it is a way of protecting your network against MAC address spoofing. The following are some benefits of port security: The following are some benefits of port security:

Preventing unauthorized access: Port security diminishes the chances of intruders getting into your network by limiting the number of devices that can be connected to a particular port.

Interference with MAC addresses: The maximum MAC address limit in the port security policy enables you to increase the number of connected devices and at the same time to prevent MAC flooding attacks.

```
S3-RF#sh po
S3-RF#sh port-security
```

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Fa0/23	2	1	0	Shutdown
Fa0/24	2	1	0	Shutdown

```
S3-RF#
```

The authentication techniques that are employed by Point-to-Point Protocol (PPP) connections are Password Authentication Protocol (PAP) and Challenge-Handshake Authentication Protocol (CHAP). The protocols give a way to verify before allowing someone or a device to the network resources. undefined

Authentication: Through the verification of the users or devices that

are allowed to connect to your network over a PPP connection, CHAP, and PAP, it is possible to avoid unauthorized access.

Secure password transmission: Both protocols, on the other hand, encrypt the passwords during the authentication process to minimize the possibility of password interception.

Network Address Translation (NAT):

Network address translation, or NAT, is a method that enables the devices in a private network to connect with the devices on the public internet. Private IP addresses are turned into public IP addresses by NAT. undefined

IP address conservation: NAT limits the use of public IP addresses, which in turn saves the address space by allowing many private IP-addressed devices to use one public IP address.

Enhanced security: Through the process of changing private IP addresses to public IP addresses, NAT hides the internal network structure and adds another security layer by blocking direct internet access to private devices.

The process of controlling the access to the network resources based on the user identity, responsibilities, or other variables is called access control. undefined

Role-based access control (RBAC): Through the process of granting permissions and access privileges to users according to the predefined user roles, RBAC makes the administration and implementation of access control policies easier to manage and apply.

ACLs, or access control lists: Rules that either allow or prohibit communication based on protocols, ports, source IP addresses, or

destination IP addresses are given by ACLs. ACLs are useful in network traffic flow management and filtering.

Redundancy

HSRP (Hot Standby Router Protocol): A gateway router can be set up in a redundant way so that it is the active gateway while the other one is in standby mode using the Hot Standby Router Protocol, or HSRP. In case the main router fails, the backup router seamlessly continues to work, thus, network connectivity is not disrupted.

advantages:

Redundancy; HSRP creates a backup gateway configuration using HSRP. When an active router fails, a standby router takes over so that there is no downtime in the network.

High Availability: a router is always available to forward traffic for the hosts that are on the same subnet as it. This increases network uptime and keeps end users connected.

Seamless Failover: HSRP provides seamless failover from the primary to the secondary router when there is a breakdown. By so doing, it minimizes packet loss and disruption to hosts connected to the network.

MN_R1

Physical Config CLI Attributes

IOS Command Line Interface

```

!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 172.16.1.68 255.255.255.192
 standby version 2
 standby 10 ip 172.16.1.126
 standby 10 priority 150
 standby 10 preempt
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 172.16.0.196 255.255.255.192
 standby version 2
 standby 20 ip 172.16.0.254
 standby 20 priority 150
 standby 20 preempt
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 172.16.0.132 255.255.255.192
 standby version 2
 standby 30 ip 172.16.0.190
 standby 30 priority 150
 standby 30 preempt
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 172.16.1.132 255.255.255.224
 standby version 2
 standby 40 ip 172.16.1.158
 standby 40 priority 150
 standby 40 preempt
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 172.16.0.4 255.255.255.128
 standby version 2
 standby 50 ip 172.16.0.126
 standby 50 priority 150
 standby 50 preempt
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 172.16.1.5 255.255.255.192
 standby version 2
 standby 60 ip 172.16.1.62
 standby 60 priority 150
 standby 60 preempt
!

```

MU_R1

Physical Config CLI Attributes

IOS Command Line Interface

```

speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.30.1.4 255.255.255.224
 standby version 2
 standby 10 ip 10.30.1.30
 standby 10 priority 150
 standby 10 preempt
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.30.0.196 255.255.255.224
 standby version 2
 standby 20 ip 10.30.0.222
 standby 20 priority 150
 standby 20 preempt
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 10.30.1.38 255.255.255.224
 standby version 2
 standby 30 ip 10.30.1.62
 standby 30 priority 150
 standby 30 preempt
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 10.30.1.68 255.255.255.224
 standby version 2
 standby 40 ip 10.30.1.94
 standby 40 priority 150
 standby 40 preempt
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.30.0.228 255.255.255.224
 standby version 2
 standby 50 ip 10.30.0.254
 standby 50 priority 150
 standby 50 preempt
!

```

MN_R2

Physical Config CLI Attributes

IOS Command Line Interface

```

speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 172.16.1.69 255.255.255.192
 standby version 2
 standby 10 ip 172.16.1.126
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 172.16.0.197 255.255.255.192
 standby version 2
 standby 20 ip 172.16.0.254
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 172.16.0.133 255.255.255.192
 standby version 2
 standby 30 ip 172.16.0.190
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 172.16.1.133 255.255.255.224
 standby version 2
 standby 40 ip 172.16.1.158
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 172.16.0.5 255.255.255.128
 standby version 2
 standby 50 ip 172.16.0.126
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 172.16.1.5 255.255.255.192
 standby version 2
 standby 60 ip 172.16.1.62
!
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 172.16.1.194 255.255.255.224
 standby version 2

```

```

speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.30.1.5 255.255.255.224
 standby version 2
 standby 10 ip 10.30.1.30
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.30.0.197 255.255.255.224
 standby version 2
 standby 20 ip 10.30.0.222
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 10.30.1.38 255.255.255.224
 standby version 2
 standby 30 ip 10.30.1.62
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 10.30.1.68 255.255.255.224
 standby version 2
 standby 40 ip 10.30.1.94
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.30.0.229 255.255.255.224
 standby version 2
 standby 50 ip 10.30.0.254
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 10.30.0.133 255.255.255.192
 standby version 2
 standby 60 ip 10.30.0.190
!
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 10.30.0.6 255.255.255.128
 standby version 2
 ..

```

```

duplex auto
speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.20.1.36 255.255.255.224
 standby version 2
 standby 10 ip 10.20.1.62
 standby 10 priority 150
 standby 10 preempt
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.20.0.132 255.255.255.192
 standby version 2
 standby 20 ip 10.20.0.190
 standby 20 priority 150
 standby 20 preempt
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 10.20.1.68 255.255.255.224
 standby version 2
 standby 30 ip 10.20.1.94
 standby 30 priority 150
 standby 30 preempt
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 10.20.1.132 255.255.255.224
 standby version 2
 standby 40 ip 10.20.1.158
 standby 40 priority 150
 standby 40 preempt
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.20.0.196 255.255.255.192
 standby version 2
 standby 50 ip 10.20.0.254
 standby 50 priority 150
 standby 50 preempt
--More--

```

```

interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.20.1.37 255.255.255.224
 standby version 2
 standby 10 ip 10.20.1.62
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.20.0.133 255.255.255.192
 standby version 2
 standby 20 ip 10.20.0.190
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 10.20.1.69 255.255.255.224
 standby version 2
 standby 30 ip 10.20.1.94
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 10.20.1.133 255.255.255.224
 standby version 2
 standby 40 ip 10.20.1.158
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.20.0.197 255.255.255.192
 standby version 2
 standby 50 ip 10.20.0.254
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 10.20.1.17 255.255.255.224
 standby version 2
 standby 60 ip 10.20.1.30
!
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 10.20.0.5 255.255.255.128
 standby version 2
 standby 70 ip 10.20.0.126
!
interface GigabitEthernet0/0.80

```

EtherChannel: Numerous physical lines between switches and routers can be connected using EtherChannel to form a single logical link which gives redundancy and more bandwidth. In the instance of a failure, traffic is immediately rerouted over the intact physical lines.

advantages:

Increased Bandwidth : Ether Channel bridges numerous links into a single unit, thereby boosting the complete bandwidth that can be used. This increased ability to move data is crucial in networks that must manage large amounts of information, such as those used by data centers or businesses running apps that demand a high bandwidth.

Load Balancing: Ether Channel uses load balancing to distribute traffic uniformly over all linked channels. This tactic avoids having a single link become a choke-point. It makes the best use of resources and boosts network efficiency while simultaneously lessening traffic jams.

Redundancy and Resilience: Ether Channel offers backup routes by providing link redundancy, enhancing fault tolerance. When one connection breaks, it smoothly shifts traffic onto other working links. This ready-plan ensures connectivity stays intact, lessening the impact brought by such breakdowns.

```

      d - default port

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)          PAgP       Fa0/4 (P) Fa0/5 (P)
S1-MN#
```

 S2-MN

Physical Config CLI Attributes

IOS Command Line In

```

      w - waiting to be aggregated
      d - default port

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)          PAgP       Fa0/4 (P) Fa0/5 (P)
S2-MN#
```

S1-MU

Physical Config CLI Attributes

IOS Command Line Inter

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Fa0/4 (P) Fa0/5 (P)

S2-MU

Physical Config CLI Attributes

IOS Command Line In

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Fa0/4 (P) Fa0/5 (P)

S1-RF

Physical Config CLI Attributes

IOS Command Line Interfac

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Fa0/4 (P) Fa0/5 (P)

S2-RF

Physical Config CLI Attributes

IOS Command Line Interfac

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Fa0/4 (P) Fa0/5 (P)

Spanning Tree Protocol (STP): To prevent loops, STP is a network protocol that cuts the unnecessary connections in Ethernet networks selectively. STP is the method that makes sure there is only one active channel between switches to stop network loops that can cause broadcast storms and other network problems.

advantages:

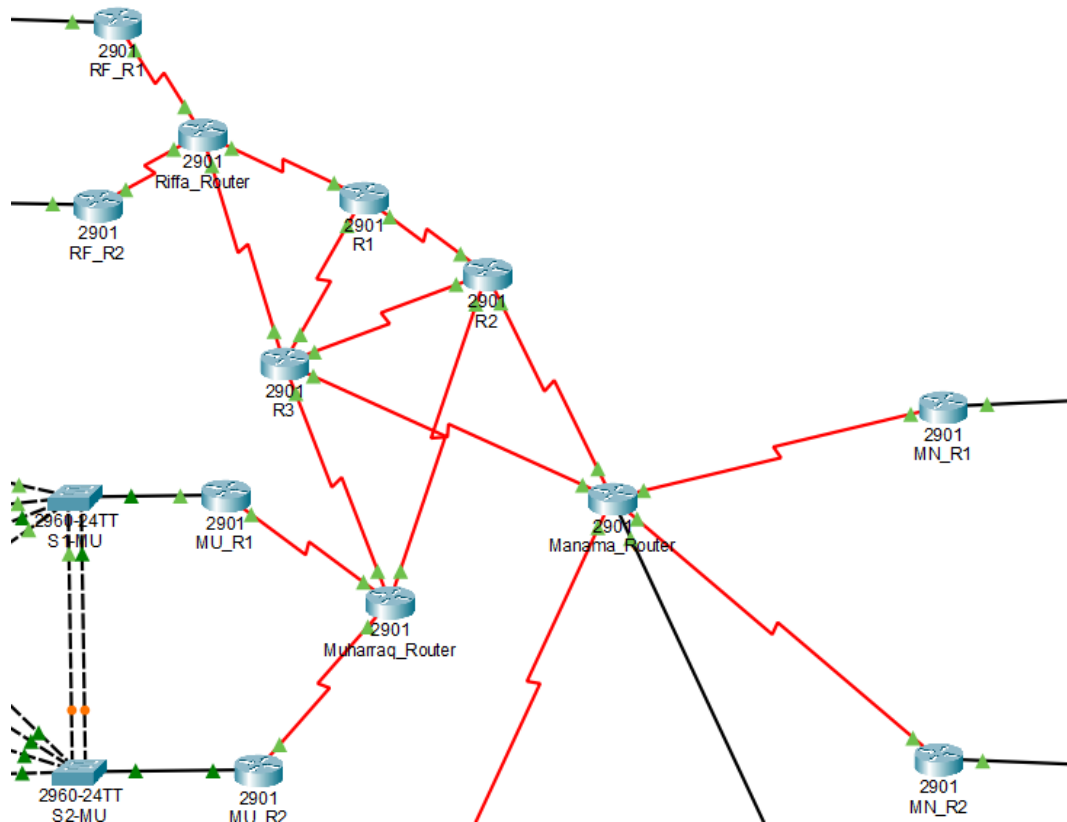
Link redundancy is provided to prevent unwanted loops

Broadcast Storms need to be avoided

Devices that are not STP-capable for example PCs, servers, routers, or hubs not connected to other switches can be connected using edge ports

Serial Links Redundancy: A redundancy may be developed between routers by setting up numerous serial connections; in the case of a link failure, traffic is automatically rerouted over the available links using dynamic routing protocols such as OSPF or EIGRP.

Implementing network redundancy by having several routers connected in series has various benefits. It increases reliability through alternatively transmitting data so that if one link fails the system does not experience downtime. Furthermore, it is able to distribute traffic uniformly over all available routes which makes good use of bandwidth thus enhancing performance.



Wireless

WPA Encryption: WPA offers encryption methods to protect data techniques such as WPA-PSK and WPA-Personal. In this mode, a pre-shared key (PSK), often known as a passphrase or password, is utilized. Every device linked to the wireless network has to have the same PIN for authentication and encryption.

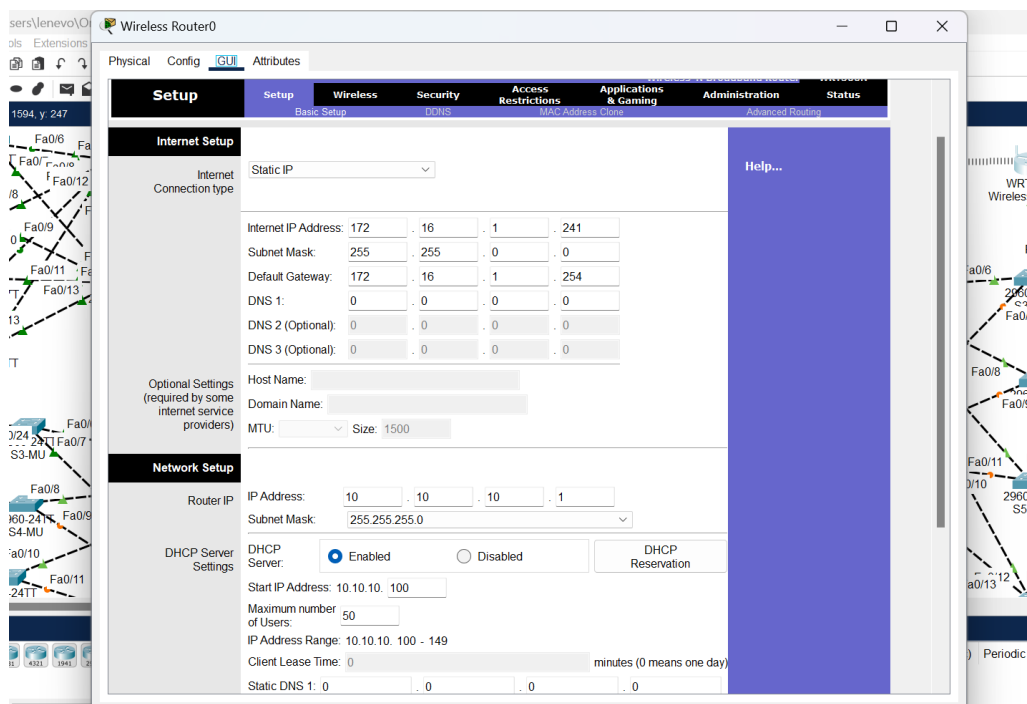
WPA-Enterprise: One example of a centralized authentication server used in this mode is RADIUS (Remote Authentication Dial-In User Service). Each user has unique login credentials that provide customized authentication and encryption.

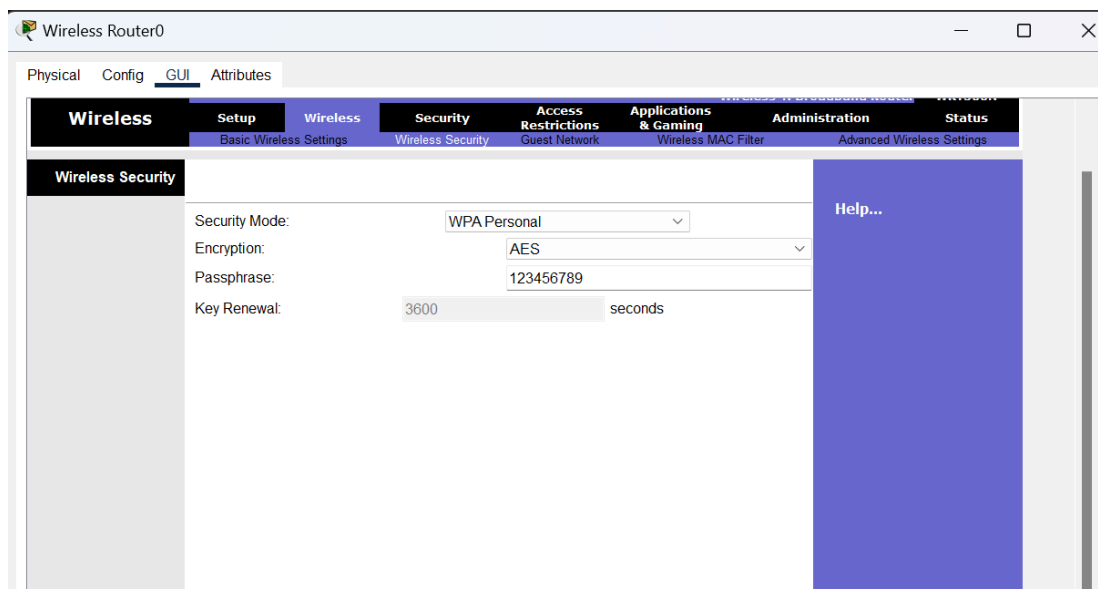
WPA2: An enhanced and more secure variant of WPA is called WPA2. Instead than using TKIP, it uses the more dependable Advanced Encryption Standard (AES) encryption technique.

WPA2 PSK (Personal) and WPA2-Enterprise have the same encryption and authentication methods as WPA.

AES (Advanced Encryption Standard) is a symmetric encryption technique that is widely used to protect data in a range of applications, including wireless networks. It operates on data blocks of a predetermined size between the parties participating in the transmission using a shared secret key. AES-256 is the maximum secure key size that it supports. There are also key sizes of 192, 256, and 128 bits. It features efficient implementations and offers a great trade-off between performance and security. When all is said and done, AES encryption provides strong and reliable data protection across wireless networks.

In the Temporal Key Integrity Protocol (TKIP), passphrases are replaced with pre-shared keys (PSKs) in the form of passphrases. The passphrase is used to produce the encryption keys using a key derivation procedure. Strong and unique passphrases are essential for increased security. When configuring TKIP security, the password is entered in the appropriate field. Increasing security requires keeping the passphrase secret and changing it often.





PPP technology

PPP is a set of communication rules that carry multiprotocol data over point-to-point connections. The system is comprised of three central sections: the packaging up of information together with a Link Control Protocol that helps in establishing connections; as well as a collection of other network control protocols used in setting up Network Layer protocols. In the OSI model, PPP is used at the Data Link Layer allowing network layer protocols to function. Essentially, PPP is a point-to-point protocol but in actual setups, a PPP server will work like a central hub for numerous.

PPP is a flexible protocol that is applied for different purposes. It can be applied to various physical media, such as dial-up modems, SSL-encrypted VPN connections, and Ethernet to exchange data between connected nodes. PPP provides point-to-point communications and can encapsulate IP and other network layer data. It was previously widely used for dial-up Internet access when customers connected to Internet service providers via a public-private partnership (PPP). PPP models, such as PPPoE and PPPoATM, are still in use today for DSL

and cable modem connections but are often done transparently by modems or routers.

Advantages of PPP:

****Strong authentication****

Mutual authentication between client and server is provided by CHAP, which reduces the possibility of unauthorized access by authenticated clients.

****High-security level****

CHAP dynamically creates challenges for each session, so it is not easy for attackers to predict or extract authentication data, thus adding an additional level of security.

****Prevention of Replay Attacks****

CHAP prevents the use of replay attacks because it uses unique challenge messages for each session, making it impossible for the attacker to make use of past responses to crack its way to unauthorized access.

****High Compatibility****

Flexibility here applies because CHAP supports many authentication modes, making it applicable to several different networking environments and systems.

****Scalability****

It is highly scalable as it can be used to accommodate various systems that can support a lot of needs. It is used in VPN connections and ensures secure access to network resources from locations where access is impossible.

****Resilience****

The ability of CHAP to continue its authentication process over time makes it very resilient where network stability is minimal.

Manama_Router

Physical Config CLI Attributes

IOS Command Line Interface

```
!  
!  
!  
!  
!  
!  
no ip cef  
no ipv6 cef  
!  
!  
!  
username Batelco_ISP password 0 cisco123  
!  
!
```

check (1) Manama_Router

Physical Config CLI Attributes

IOS Command Line Interface

```
!  
interface Serial0/1/1  
 ip address 193.200.35.17 255.255.255.252  
 encapsulation ppp  
 ppp authentication chap  
 ip nat outside  
 clock rate 2000000  
!
```

(1) Batelco_ISP

Physical Config CLI Attributes

IOS Command Line Interface

```
!  
!  
username Manama_Router password 0 cisco123  
username P1 password 0 P1PASS  
username P2 password 0 P2PASS  
!  
!
```

k (1) Batelco_ISP

Physical Config CLI Attributes

IOS Command Line Interface

```
!  
interface Serial0/1/1  
 ip address 193.200.35.18 255.255.255.252  
 encapsulation ppp  
 ppp authentication chap  
!  
!
```

PPPoE technology

Point-to-Point Protocol over Ethernet (PPPoE) is a network protocol enabling communication between the endpoints of a network by encapsulating Point-to-Point Protocol (PPP) frames inside an Ethernet frame. It is used by internet service providers (ISP) for the delivery of internet access services, especially over broadband connections. The great majority of their DSL services use PPPoE. It allows the ISPs to manage multitudes of client systems, their authentication of access, and to track customer data usage. It supports services for both data encryption and data compression. PPPoE was designed to overcome deficiencies in PPP and to provide a solution for organizations using common customer premises equipment in multihost environments.

PPPoE Conference Schedule:

The user's device initiates a PPPoE session by sending a PPPoE discovery cycle with at least information such as the user's username and password

The ISP's access concentrator, typically a DSL modem or router, undergoes a discovery cycle and sends back a PPPoE session confirmation.

PPPoE Session License:

The user device, upon receiving the session confirmation, sends an authorization request to the concentrator containing the authentication details.

The ISP's access concentrator will check the returned credentials, after which an authorization response will be issued.

PPPoE Conference Schedule:

When a license is created, a PPPoE session is created between the user's device and the ISP's access point.

This session acts as a virtual point-to-point connection over Ethernet.

PPP connection:

Once a PPPoE session is established, PPP frames can be sent back and forth from the user's device to the ISP's access concentrator.

Control messages in PPP frames.

Advantages of implementation of PPPoE:

**** Secure Connection **** It facilitates authentication for increased network safety.

**** Broadband Flexibility **** It accommodates several broadband technologies, including DSL.

**** Efficient Bandwidth Usage**** It makes use of borrowed bandwidth in an effective manner through session-based traffic management.

**** Easy Configuration**** It is easy to configure since a user can simply configure the DSL modems or routers.

****Remote Network Access**** It is easy to connect to the HQ network.

**** Scalability**** It is easy to scale up and accommodate an increased number of telecommuters.

-- FOR EVERY TIME TRY TO CONNECT THE P1+P2 TO THE ISP --

PC1

Username: P1

Pass: P1PASS

PC2

Username: P2

PassLP2PASS

The image shows a network diagram on the left and a command prompt window on the right. The diagram illustrates a switch configuration with ports Fa0/5, Fa0/11, Fa0/12, Fa0/13, and Fa0/24. It includes a 2960-24TT switch and a 2960-24TT switch. The command prompt window displays the configuration for the switch, including the physical address, link-local IPv6 address, IPv4 address, subnet mask, default gateway, DHCP servers, and DHCPv6 client DUID. The configuration is for a switch named 'S2-MU'.

```

Physical Config Desktop Programming Attributes

Command Prompt
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-6C-74-32-09-00-01-43-C4-26-84
DNS Servers.....: ::
0.0.0.0

Dialer1 Connection:

Connection-specific DNS Suffix...: 0001.C960.D3E9
Physical Address.....: FE80::201:43FF:FEC4:2684
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 193.200.35.11
Subnet Mask.....: 255.255.255.255
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-6C-74-32-09-00-01-43-C4-26-84
DNS Servers.....: ::
0.0.0.0

Virtual-Access1 Connection:

Connection-specific DNS Suffix...: 0001.64EE.3C2E
Physical Address.....:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-6C-74-32-09-00-01-43-C4-26-84
DNS Servers.....: ::
0.0.0.0

Virtual-Access2 Connection:
--More--

```

The image shows a network diagram on the left and a command prompt window on the right. The diagram illustrates a switch configuration with ports Fa0/5, Fa0/11, Fa0/12, Fa0/13, and Fa0/24. It includes a 2960-24TT switch and a 2960-24TT switch. The command prompt window displays the configuration for the switch, including the physical address, link-local IPv6 address, IPv4 address, subnet mask, default gateway, DHCP servers, and DHCPv6 client DUID. The configuration is for a switch named 'S2-MU'.

```

Physical Config Desktop Programming Attributes

Command Prompt
Connection-specific DNS Suffix...: 00E0.F75D.8D63
Physical Address.....: FE80::2E0:8FFF:FEAD:64A1
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-6E-0C-6A-94-00-E0-8F-AD-64-A1
DNS Servers.....: ::
0.0.0.0

Dialer1 Connection:

Connection-specific DNS Suffix...: 0090.2B06.D0AA
Physical Address.....: FE80::2E0:8FFF:FEAD:64A1
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 193.200.35.12
Subnet Mask.....: 255.255.255.255
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-6E-0C-6A-94-00-E0-8F-AD-64-A1
DNS Servers.....: ::
0.0.0.0

Virtual-Access1 Connection:

Connection-specific DNS Suffix...: 0001.9696.158D
Physical Address.....:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-6E-0C-6A-94-00-E0-8F-AD-64-A1
DNS Servers.....: ::
0.0.0.0

```

- NAT/PAT

NAT:

Network Addressing Translation is a technology used to give 2 IP addresses for the device. One of the IP addresses is referred to in the packet tracer as inside local which is used for communication internally and the other is referred to as inside global which is used for outside communication. The main goal of Nat is to save public IPV4 addresses since they are limited.

There are 3 types of NAT:

1. Static NAT:

This type allows us to sign each internal address with an external IP address for external communication, keeping it fixed using one-to-one mapping. The internal IP address will always translate to the same external IP address. But one disadvantage of this type is that you must assign the external IP address manually to each internal IP address which is not practical for large-scale networks. It is primarily used to provide reliable access to specific external networks, like web and email servers. You can access the servers from any outside device in any branch using the new external IP address, here we access the servers using devices from all the 3 different branches.

2. Dynamic NAT:

The external IP addresses are assigned using a different method with this kind of NAT. It is necessary to establish an IP address pool with a beginning IP address and an ending IP address. when establishing a connection to a network that is external. Any external IP address in the pool that is available is automatically provided for translation by dynamic NAT. The mapping is dynamic, changing with each new connection. Due

to its greater scalability, this type is utilized for general internet access.

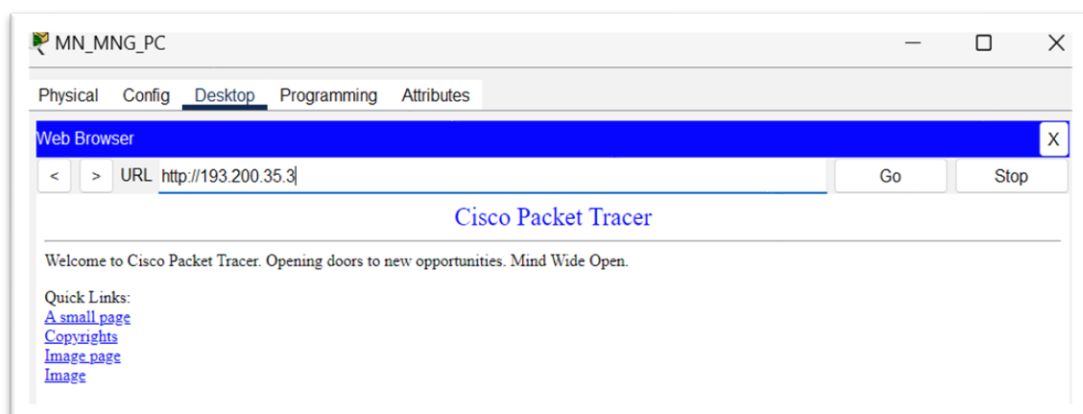
3. Port Address Translation (PAT):

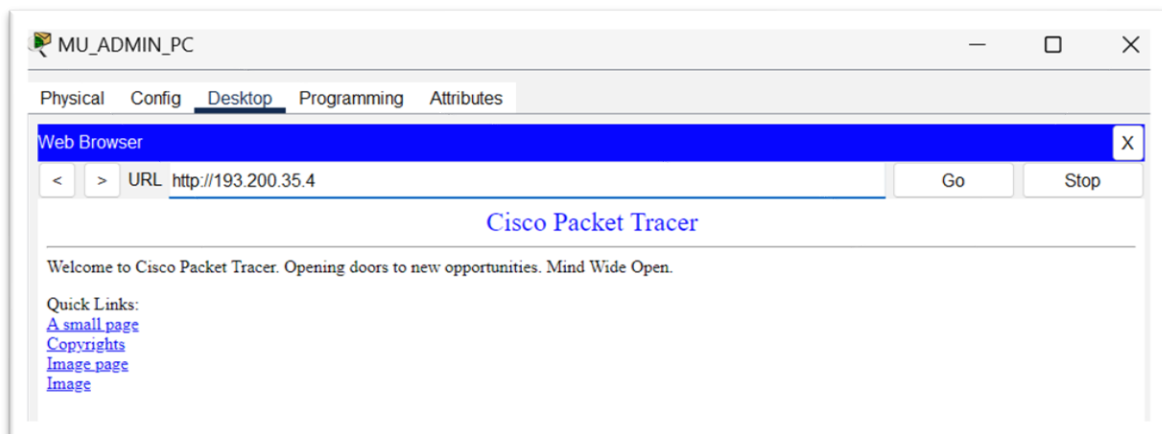
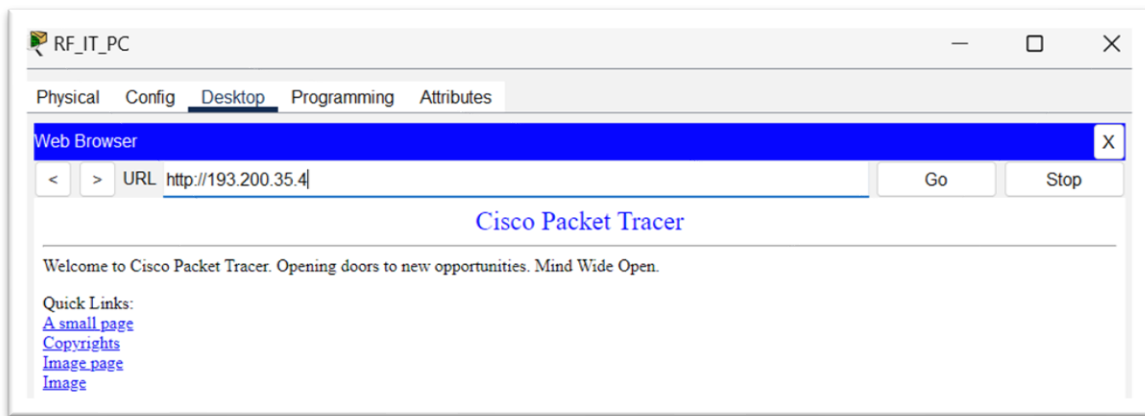
This type of overloading NAT permits port numbers to let several devices share a single public IP address. The devices can communicate with external networks using the same public IP address as long as different port numbers are being used, as it translates both the IP address and the port number. It is typically utilized in scenarios where the number of available IP addresses is restricted, such as in cases of work from home or small offices with constrained networks.

In the packet tracer, we implemented the Nat in two ways:

- First apply the first type of NAT the static NAT on the Manama router for the two available servers we have in our topology which are the email server and the web server. This was done by assigning manually two public IP addresses to the two servers we had. You can access the servers from any outside device in any branch using the new external IP address, here we access the servers using devices from all the 3 different branches.

```
ip nat inside source static 172.16.2.26 193.200.35.3  
ip nat inside source static 172.16.2.27 193.200.35.4
```





```
Manama_Router#sh ip nat tr
Pro  Inside global      Inside local      Outside local     Outside global
---  193.200.35.3         172.16.2.26      ---              ---
---  193.200.35.4         172.16.2.27      ---              ---
tcp  193.200.35.3:80      172.16.2.26:80   172.16.1.65:1025 172.16.1.65:1025
tcp  193.200.35.4:80      172.16.2.27:80   10.20.1.1:1025   10.20.1.1:1025
tcp  193.200.35.4:80      172.16.2.27:80   10.30.0.193:1025 10.30.0.193:1025
```

- Secondly, we applied the third type of NAT, the PAT on the rest of the devices by creating an IP address pool containing only two unused IP addresses so that we can take advantage of the port numbers. Also, creating an access list in which we permitted all the IP addresses of the devices we need to have access to the internet and entering the overload command except the financial and HR pcs because they should not have access to the internet. So, we permit the devices we need, and the ACL by default denies the rest of the devices.

```
ip access-list standard PAT
permit 172.16.1.64 0.0.0.63
permit 172.16.0.192 0.0.0.63
permit 172.16.0.128 0.0.0.63
permit 172.16.1.0 0.0.0.63
permit 172.16.1.160 0.0.0.31
permit 172.16.1.192 0.0.0.31
permit 172.16.1.240 0.0.0.15
permit 10.30.1.0 0.0.0.31
permit 10.30.0.192 0.0.0.31
permit 10.30.1.32 0.0.0.31
permit 10.30.0.128 0.0.0.63
permit 10.30.0.0 0.0.0.127
permit 10.30.1.96 0.0.0.15
permit 10.20.1.32 0.0.0.31
permit 10.20.0.128 0.0.0.63
permit 10.20.1.64 0.0.0.31
permit 10.20.1.0 0.0.0.31
permit 10.20.0.0 0.0.0.127
permit 10.20.1.96 0.0.0.31
permit 193.200.35.0 0.0.0.7
permit 172.16.1.192 0.0.0.15
```

```
ip nat pool PAT 193.200.35.5 193.200.35.6 netmask 255.255.255.248
ip nat inside source list PAT pool PAT overload
```

- Access List (ACL)

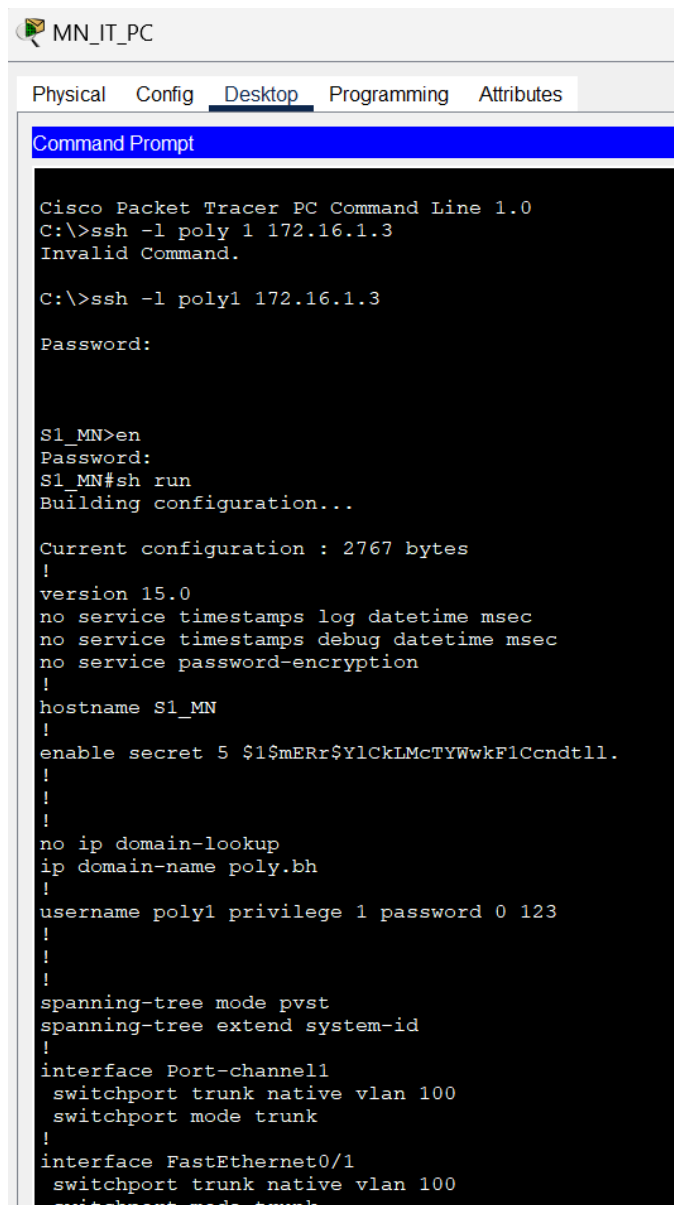
There are two types of ACL

- 1- Standard ACL: This is the most basic type of ACL using only source and destination IP addresses, it is used to grant access permission for a user or group to a specific file or device and we can apply it to any interface.
- 2- Extended ACL: A more complex type of ACL. In this ACL in addition to port and destination IP address, you can set extra parameters like protocols, ports, TCP, and ICMP traffic.

In this project, we had 4 requirements regarding ACL:

1. In the first requirement we used SSH to grant access to only the IT staff to the network devices (switches and routers). This was done by creating a crypto key using RSA then assigning a username and password (123) for each device and assigning an ip address to VLAN 60 (IT) for every device. Then we did a normal ACL to permit only the IT staff of all branches to be able to access the devices. And we can check it from the command prompt of the pcs.

Here in the first pic, we can access the network because we are on an IT PC but in the second we can't because we are on a different PC that is not an IT branch, we can even access it from a different branch using an IT branch PC:



MN_IT_PC

Physical Config Desktop Programming Attributes

Command Prompt

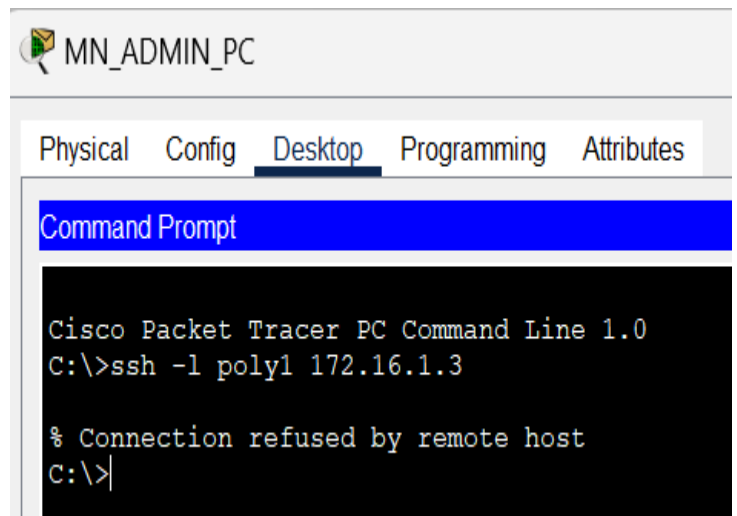
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l poly 1 172.16.1.3
Invalid Command.

C:\>ssh -l poly1 172.16.1.3

Password:

S1_MN>en
Password:
S1_MN#sh run
Building configuration...

Current configuration : 2767 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1_MN
!
enable secret 5 $1$mERr$YlCkLMcTYWwkF1Cndt11.
!
!
no ip domain-lookup
ip domain-name poly.bh
!
username poly1 privilege 1 password 0 123
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
 switchport trunk native vlan 100
 switchport mode trunk
!
interface FastEthernet0/1
 switchport trunk native vlan 100
 switchport mode trunk
```



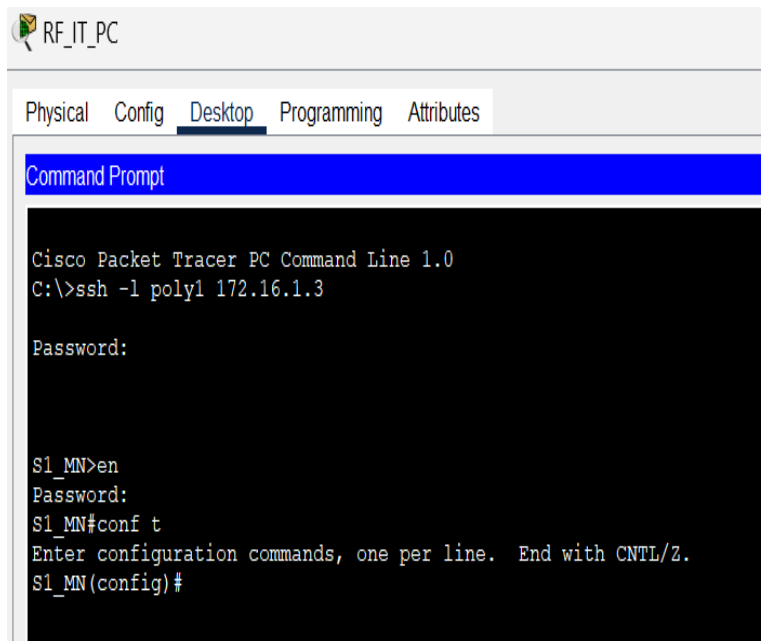
MN_ADMIN_PC

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l poly1 172.16.1.3

% Connection refused by remote host
C:\>|
```



RF_IT_PC

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l poly1 172.16.1.3

Password:

S1_MN>en
Password:
S1_MN#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1_MN(config)#
```

2. In the second requirement we used an extended ACL to block all ICMP traffic to our servers and we must address the servers by their external IP from NAT.

```
permit 10.20.1.30 0.0.0.0  
access-list 110 deny icmp any host 193.200.35.4  
access-list 110 deny icmp any host 193.200.35.3
```

3. For the third requirement we used a standard ACL so we can permit all the pcs to have access to the internet except the HR and Finance departments and applied it to the serial int connecting the Manama router to the Batelco router.

```
ip access-list standard PAT
permit 172.16.1.64 0.0.0.63
permit 172.16.0.192 0.0.0.63
permit 172.16.0.128 0.0.0.63
permit 172.16.1.0 0.0.0.63
permit 172.16.1.160 0.0.0.31
permit 172.16.1.192 0.0.0.31
permit 172.16.1.240 0.0.0.15
permit 10.30.1.0 0.0.0.31
permit 10.30.0.192 0.0.0.31
permit 10.30.1.32 0.0.0.31
permit 10.30.0.128 0.0.0.63
permit 10.30.0.0 0.0.0.127
permit 10.30.1.96 0.0.0.15
permit 10.20.1.32 0.0.0.31
permit 10.20.0.128 0.0.0.63
permit 10.20.1.64 0.0.0.31
permit 10.20.1.0 0.0.0.31
permit 10.20.0.0 0.0.0.127
permit 10.20.1.96 0.0.0.31
permit 193.200.35.0 0.0.0.7
permit 172.16.1.192 0.0.0.15
```

```
interface Serial0/1/1
ip address 193.200.35.17 255.255.255.252
encapsulation ppp
ppp authentication chap
ip access-group PAT out
ip nat outside
clock rate 2000000
```

4. In the fourth requirement we used an extended ACL to allow internet to access to the servers for only email and internet traffic only and block access to the remaining parts of the network.

```
ip access-list extended ALLOW_EMAIL_WEB_ESTABLISHED
permit tcp any any eq smtp
permit tcp any any eq 587
permit tcp any any eq 465
permit tcp any any eq pop3
permit tcp any any eq 995
permit tcp any any eq 143
permit tcp any any eq 993
permit tcp any any eq www
permit tcp any any eq 443
permit tcp any any established
```

Port numbers:

587: SMTP specifically created for clients to submit emails.

465: SMTPS encrypted email transmission

995: pop3s encrypted connection for receiving emails using pop3 protocol

143: IMAP which serves as a substitute for pop3

993: IMAPS which is an IMAP with more security using encryption

443: HTTPS provides an encrypted connection to the web

Established: Refers to packets that match and are already a part of a TCP connection

Conclusion

To sum up, a strong network infrastructure is indispensable to driving GUB's growth as it moves to Bahrain and opens branches in both Riffa and Muharraq. The main focus will be on scalability, repeatability, availability and security as they are the essential components of successful communications between all domains. Having the right set of technologies, conducting regular inspections, and following established good practices means that the bank will have a strong network infrastructure that is geared towards its goals and will foster growth.

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